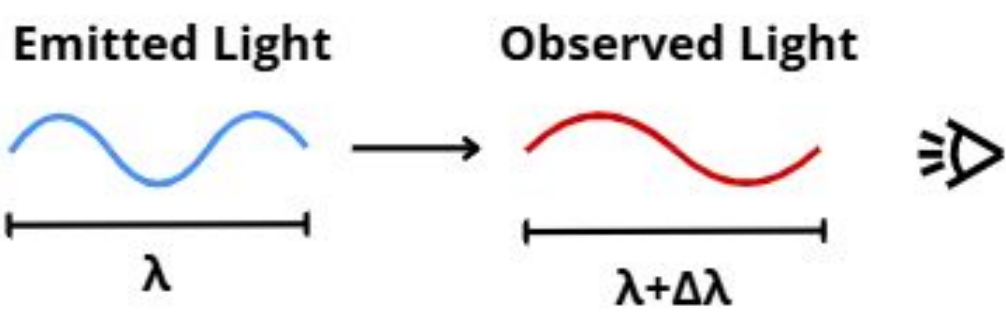


Combining datasets with different ground truths using Low-Rank Adaptation to generalize image-based CNN models for photometric redshift prediction



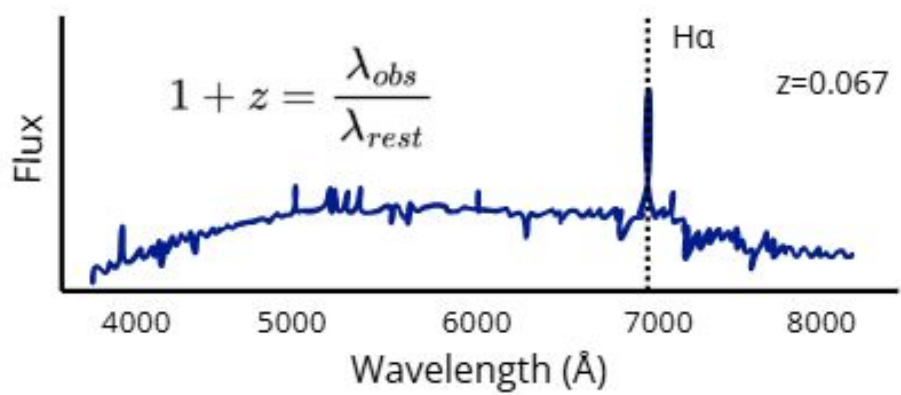
How can we bridge the redshift gap for upcoming massive surveys?

What are redshifts?



- Redshift is the stretching of electromagnetic waves as a source moves away from an observer.
- In cosmology, redshifts occur as space expands. Redshifts are vital for mapping cosmic structure and understanding dark energy and dark matter.

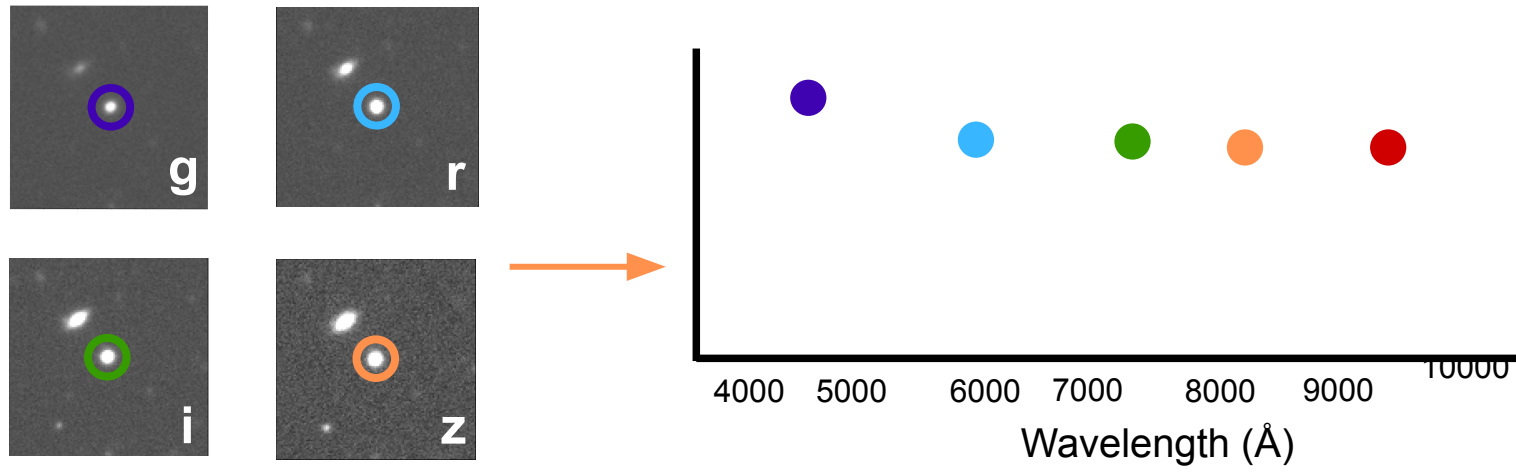
Spectroscopy



Credit: The
2dF Galaxy
Redshift
Survey and
CSIRO

- Light from a source is distributed across a spectrum. Astronomers compare spectral emission lines to their rest frame wavelengths to obtain spec-z.
- They are precise but are impractical for large scale surveys such as LSST or Euclid that will look at billions of galaxies.

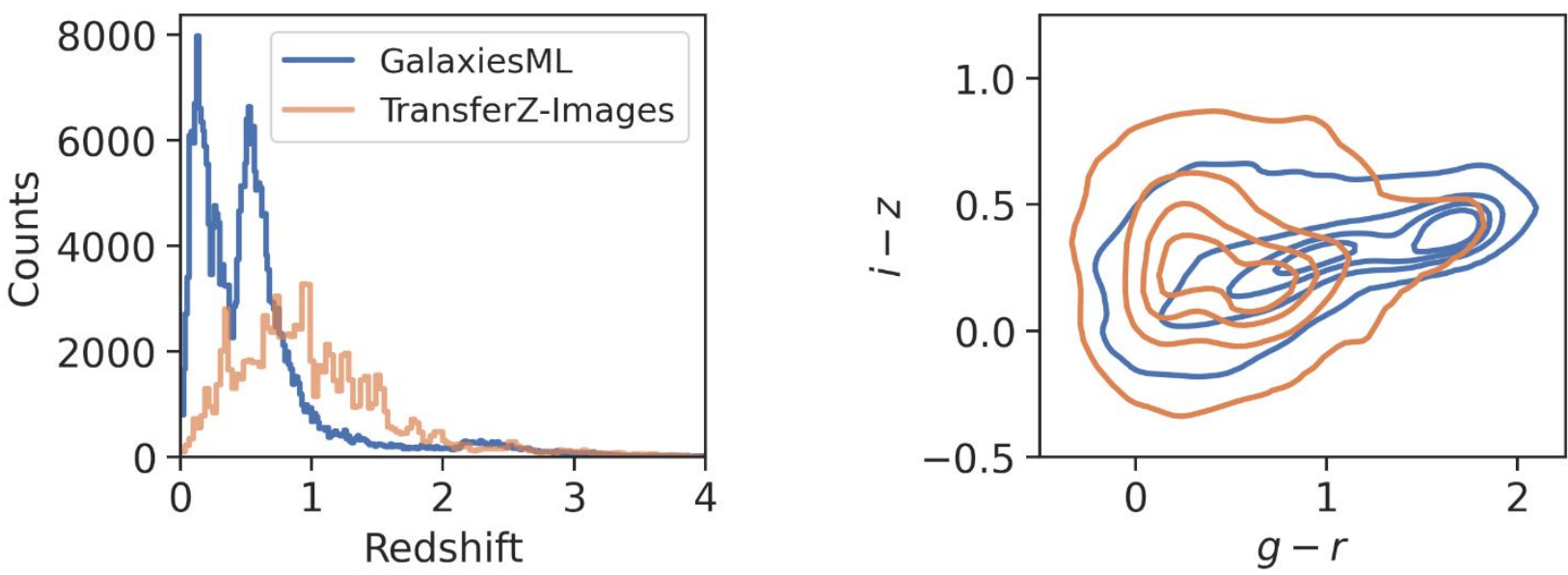
Photometry



- Galaxy brightness (photometry) is measured from images at different wavelength bands.
- The band-photometry is compared to galaxy templates at different redshifts to obtain photo-z.
- Enables analysis on larger datasets at a cost to precision

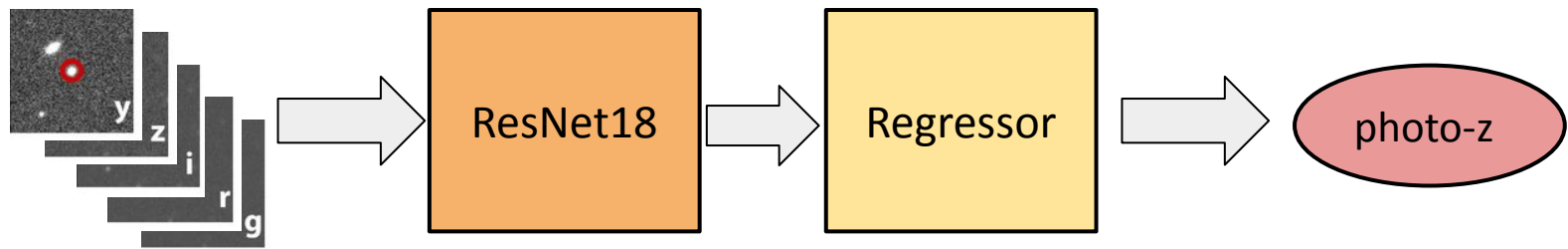
How can we combine diverse sources of ground truth to yield more generalizable redshift prediction models?

Data



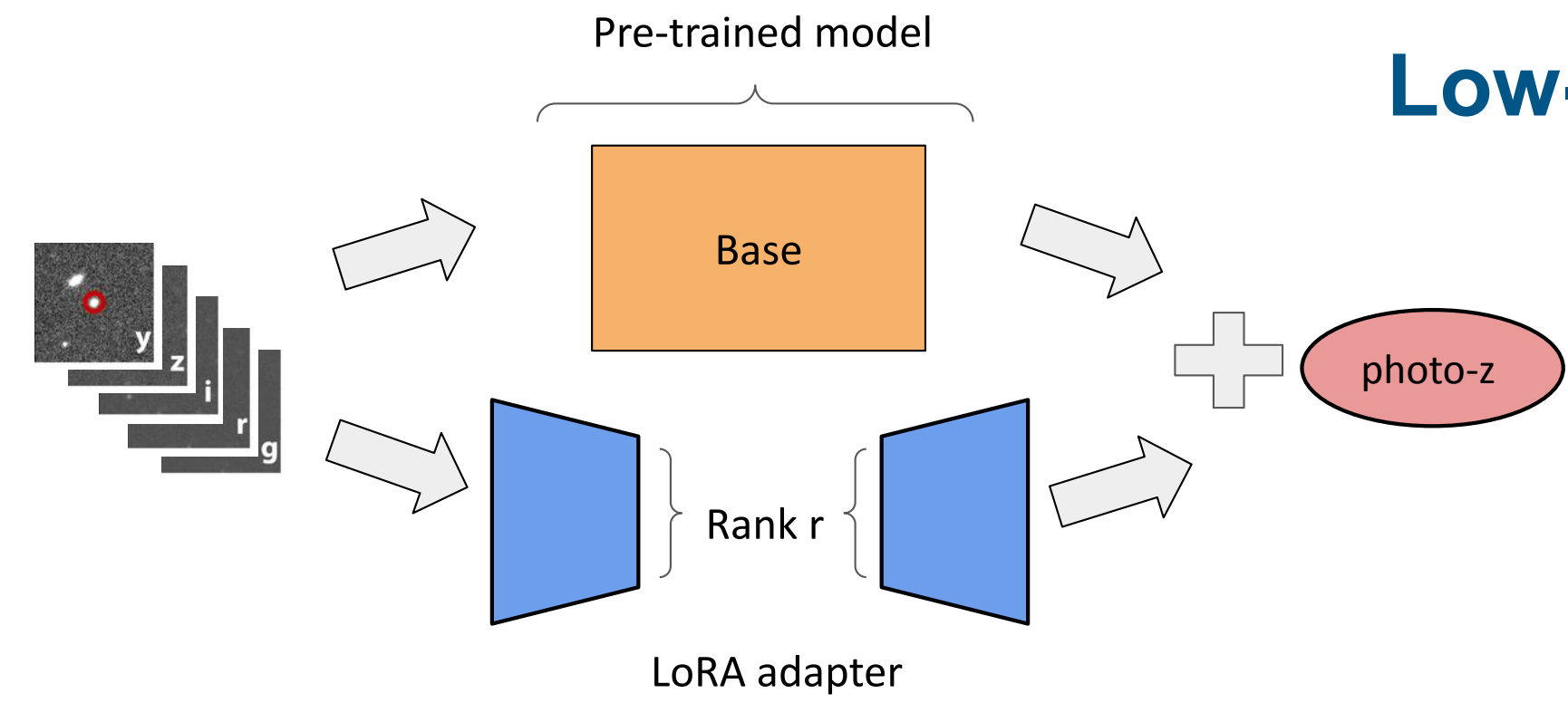
We use TransferZ-Images, with photometric redshift ground truths derived from 35 filters and GalaxiesML with spectroscopic redshift ground truths. The two datasets have different redshift distributions (Left) and color distributions (Right), that complement each other. Together, there are ~400,000 galaxies with images in five filters: g, r, i, z, y; large scale surveys will produce photometry in five filters.

Model



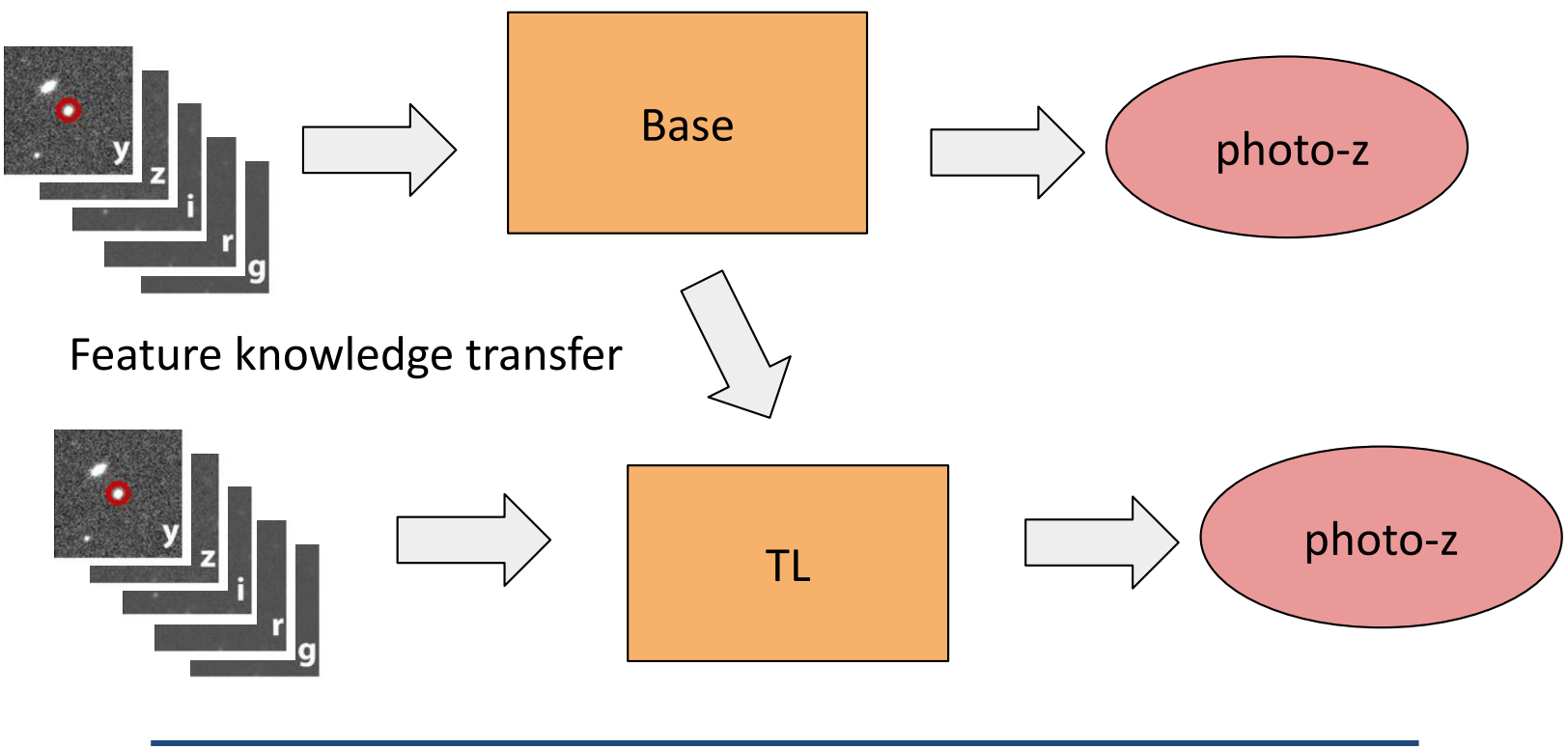
Our model consists of a ResNet18 backbone followed by a regressor with two linear layers with 512 and 256 neurons, with dropout layers (dropout=0.5) and SiLU activation between them.

Low-rank adaptation



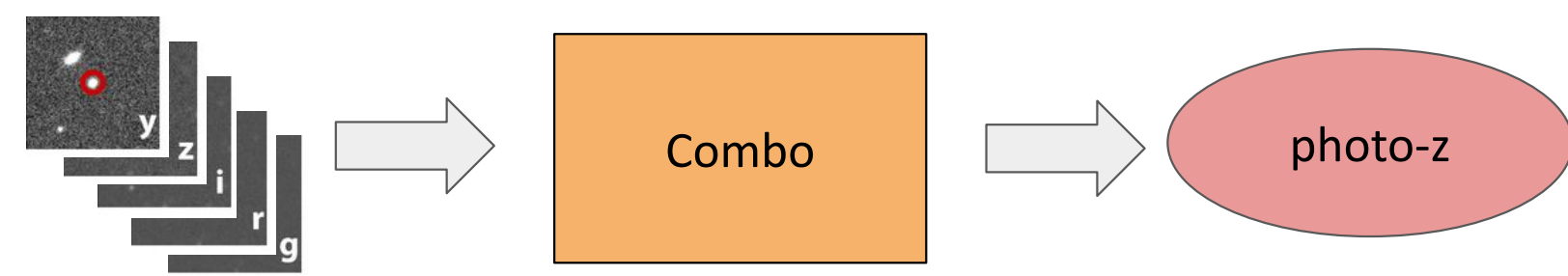
A baseline model is trained on TransferZ-Images. Next, low-rank matrices are inserted into the model's weights that are trained on GalaxiesML.

Transfer Learning



A baseline model is trained on TransferZ-Images. The model weights are frozen except in the input and regressor layers, and the model is fine-tuned with GalaxiesML.

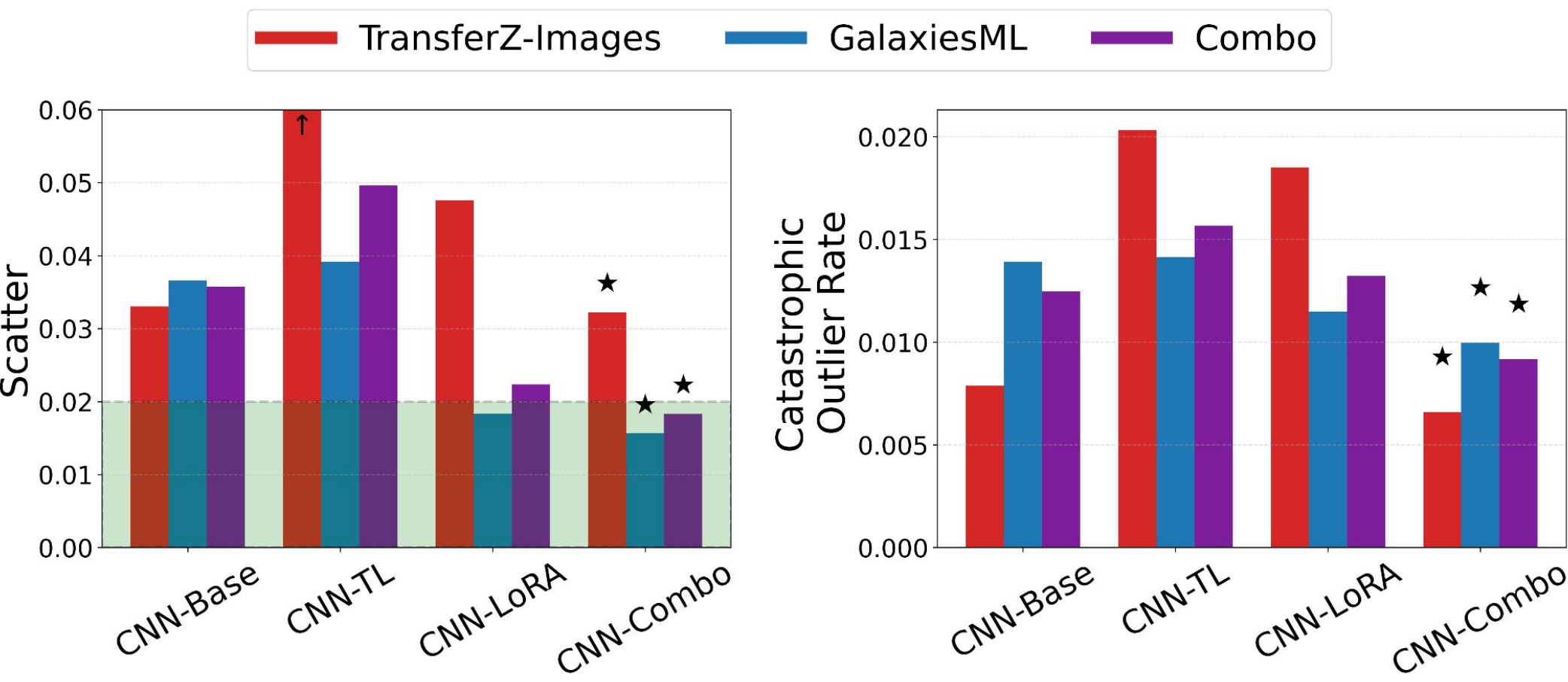
Combined training



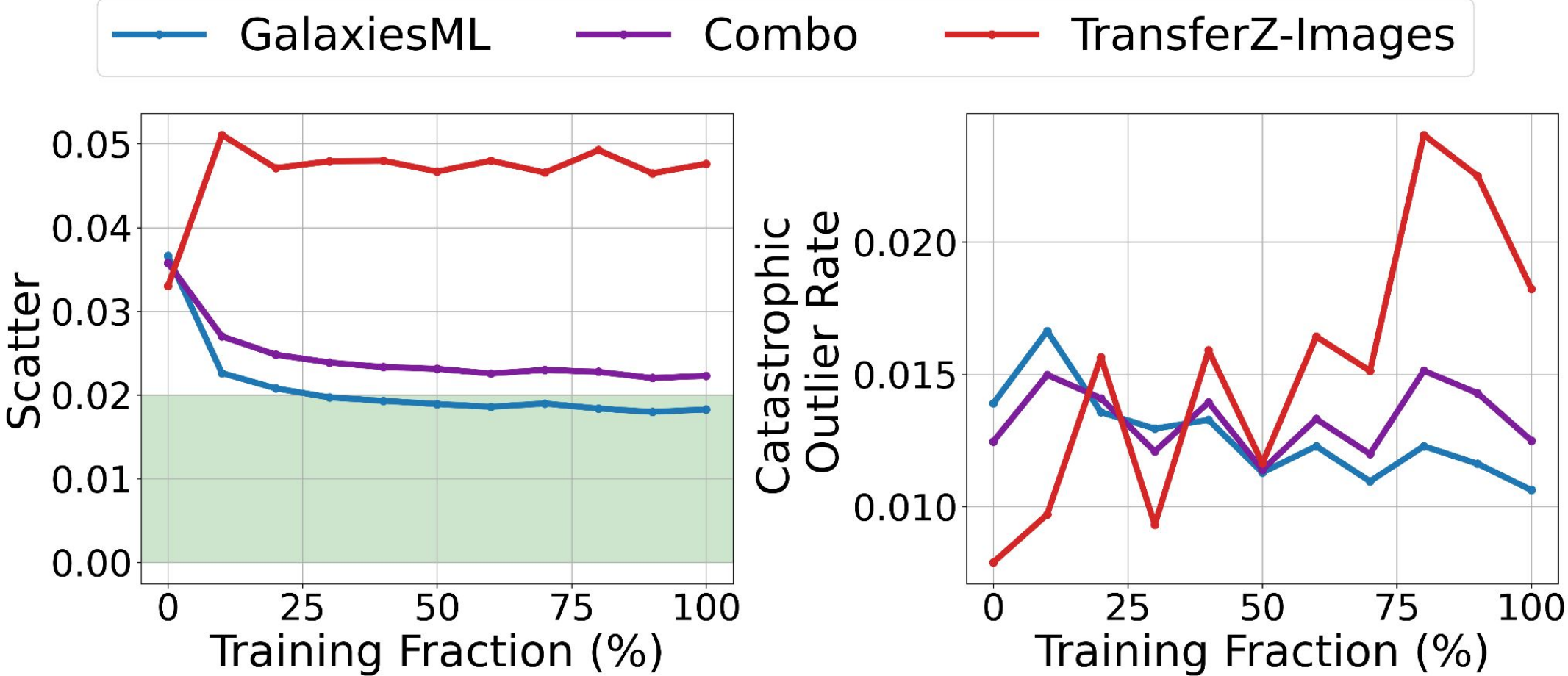
The model is trained on a dataset called Combo, that combines TransferZ-Images and GalaxiesML

LoRA provides a middle ground between full retraining and zero-shot generalization.

LoRA can be trained with small data fractions without much degradation in accuracy.



LoRA performs better than transfer learning, but not as good as retraining on combined data.



The LoRA model performance does not improve significantly beyond using 40-50% of GalaxiesML for fine-tuning.

LoRA displays potential as a way to integrate diverse ground truths, and leverage pre-existing models with new data.

$$\sigma_{NMAD} = 1.48 \times MAD|\Delta z| \quad \sigma_c = f(|z_{pred} - z_{truth}| > 1)$$